UNIT II – Lathe, Drilling, Boring, Shaping, Slotting & Planing

23A03501 MACHINING PROCESSES · B.Tech Mech · JNTUK

I. Engine Lathe – Principle & Working

Lathe = machine tool that removes metal from a **rotating** workpiece using a **single-point** cutting tool moving linearly (parallel / perpendicular to axis).

- Work → rotates (primary cutting motion) ; tool → feed motion
- Produces **cylindrical / conical / flat / threaded** surfaces
- **Engine lathe** (centre lathe) = most common; named so as early ones were driven by steam engines
- Power from motor → belt/gears → headstock spindle → work
- Feed from spindle → gear train → **feed rod** (turning) or **lead screw** (threads)

Main parts (Fig 1)

- **Bed** – C.I. base, with **V / flat guideways**; carries all parts; rigid, damps vibration
- **Headstock** – fixed at left; has hollow **spindle**, speed gearbox, Morse taper; holds chuck
- **Tailstock** – right, slides on bed; supports long work with **dead centre**; holds drill, reamer
- **Carriage** = saddle + cross-slide + compound rest + tool post + apron
  - **Saddle** – slides on bed along Z
  - **Cross-slide** – on saddle, moves ⊥ to axis (X); gives depth of cut / facing
  - **Compound rest** – swivels on cross slide; taper turning; short feed
  - **Apron** – hangs from saddle; feed clutches, half-nut, handwheels
- **Lead screw** – ACME thread, only for thread cutting (half nut engaged)
- **Feed rod** – plain shaft with keyway; gives power feed for turning, facing

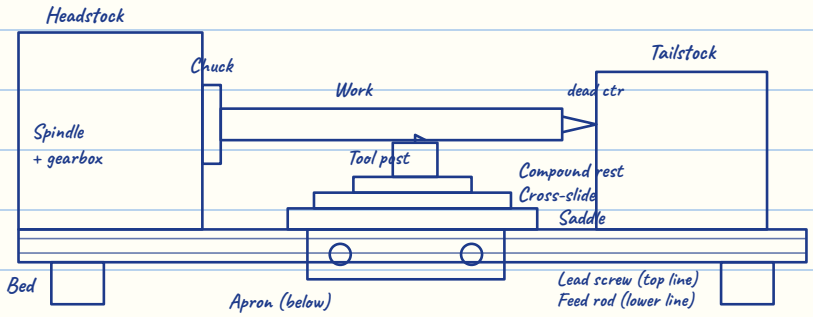


Fig 1: Engine lathe – main parts

2. Specification of a Lathe

- **Swing** over bed = max dia of work that can rotate over bed (= 2 × centre height)
- **Swing over carriage** (cross-slide) – smaller than over bed
- **Max distance between centres** = length of work between centres
- **Bed length** (overall) ; **Spindle bore** ; spindle nose type
- Range of spindle speeds (rpm) & feeds ; thread pitch range
- Motor power (kW) ; Morse taper no. of tailstock ; lead screw pitch
- Eg: **"300 mm × 1500 mm"** = swing 300, centre distance 1500

3. Types of Lathes

Type	Features / use
Speed lathe	Only headstock, tailstock, tool post on bed; 1200-3600 rpm; no feed gearbox; wood turning, spinning, polishing, centring
Engine / centre	General purpose; gear-driven; power feed + threads; medium production, jobs
Bench lathe	Small, mounted on bench; precision small parts, watches, instruments
Tool room lathe	Engine lathe + extra accuracy, wide speed range, taper attachment, collet; tool & die making
Capstan / Turret	Hexagonal turret (tailstock side) with 6 tools set in sequence; repetitive batch work. Capstan: turret on ram – light jobs, bar stock; Turret: turret on saddle – heavy, large jobs
Automatic	All tool movements by cams, no operator; single/multi spindle (screw machine, Swiss type); mass production
CNC lathe	Programmed (G & M codes), servo motors, turret of 8-12 tools; high accuracy, flexible; batch & mass

★ Important

- Capstan vs turret: capstan turret is on a **ram** (slides on saddle), turret lathe turret is on **saddle** (more rigid)
- Engine lathe = single tool at a time; turret = many tools in one setup → saves time

4. Work Holding Devices

(a) Chucks

- **3-jaw (self-centring / universal) chuck** – jaws move together by scroll; for round, hexagonal bars; quick; accuracy low
- **4-jaw (independent) chuck** – each jaw separate by screw; square, irregular, eccentric jobs; heavy duty; slow setting
- **Collet chuck** – split spring sleeve grips small bar; very accurate, bar work
- **Magnetic chuck** – holds thin ferrous parts by magnetic force, no distortion; used in grinding
- **Combination chuck** – jaws can move together or independently

(b) Others

- **Centres** – live (in headstock, rotates) & dead (tailstock; hardened, lubricated) ; 60° point; Morse taper shank
- **Face plate** – C.I. disc with slots; clamp irregular jobs by bolts, angle plate; balance with counterweight
- **Mandrel** – job with bored hole held between centres to turn outside (plain, gang, taper, expanding, stepped)
- **Lathe dog + driving plate** – drives work between centres
- **Steady rest** (fixed on bed, 3 jaws) – supports long slender job at middle / end; avoids deflection
- **Follow rest** – fixed on carriage, moves with tool; supports work against tool thrust (2 jaws); long thin shafts, threading

5. Tool Holders & Cutting Tools

- Tool held in **tool post** on compound rest (clamped by set screws; packing for height)
- Types of tool post: **single-screw (round)**, **four-bolt (square)**, **four-way turret**, **quick-change (Dickson / Multifix)**
- **Straight, left-bent, right-bent** tool holders → for roughing, left/right cuts

- **Boring bar holder** ; **Parting tool holder** ; **Threading tool holder** (spring type)
- Tool types: solid HSS; brazed carbide tip; **indexable insert** clamped on shank (clamped, not resharpened)
- Tip at **centre height** – else facing leaves a pip; threading angle wrong

6. Lathe Operations

Operation	Description
Straight turning	Reduce dia, tool ∥ axis
Facing	Flat face ⊥ axis; cross-slide feed
Taper turning	Conical surface
Thread cutting	Helical groove; tool feed = pitch/rev
Knurling	Diamond / straight pattern pressed by knurling wheels for grip (no chips)
Drilling	Drill in tailstock; work rotates
Boring	Enlarge existing hole by single point tool
Reaming	Finishing hole with reamer
Parting / cut-off	Narrow groove till job separated
Grooving / Chamfering	Form groove / bevel edge
Forming	Form tool gives contour in one plunge
Filing, polishing, spinning, eccentric turning	Minor ops

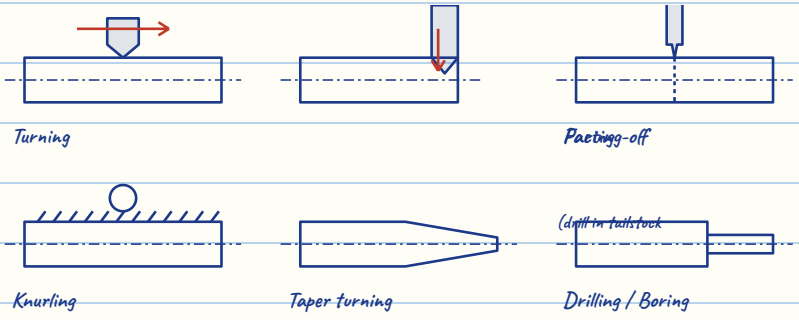


Fig 3: Basic lathe operations (schematic)

7. Taper Turning

Taper = ratio of diff in diameters to length. Conicity $K = (D - d)/l$; **half angle** α (semi-taper)

$$\tan \alpha = (D - d) / 2l$$

- D = large dia, d = small dia, l = length of tapered portion (mm)

- Taper per foot = $12(D-d)/l$; Morse taper ≈ 1 in 20 (≈ 0.05 mm/mm)

Methods

1. **Form tool** – broad nose tool ground to taper angle, plunged in; short tapers (chamfers), ≤ 10-15 mm; fast, rigid needed
2. **Compound rest swivelling** – rest set to α; hand feed by compound screw; steep tapers, short length (limited by travel); any angle
3. **Tailstock set-over** – tailstock shifted $\perp$ to axis; work axis inclined; long, gentle tapers (≤ 8-10°); centre holes get distorted; no internal tapers
4. **Taper turning attachment** – guide bar set at α on bracket behind bed; slide block moves cross-slide; accurate, long tapers (≤ 10-12°), power feed; internal too; no need to disturb tailstock
5. **Combining feeds** (CNC / hydraulic copying) – longitudinal + cross feeds simultaneously

$$\text{Tailstock set-over } S = L\,(D - d) \,/\, 2l = L \tan \alpha$$

- L = total length of work between centres, l = length of taper. If whole length tapered ($L = l$): $S = (D - d)/2$
- Alternate: $S = L \times (\text{taper ratio}) / 2$, with taper = $(D-d)/l$

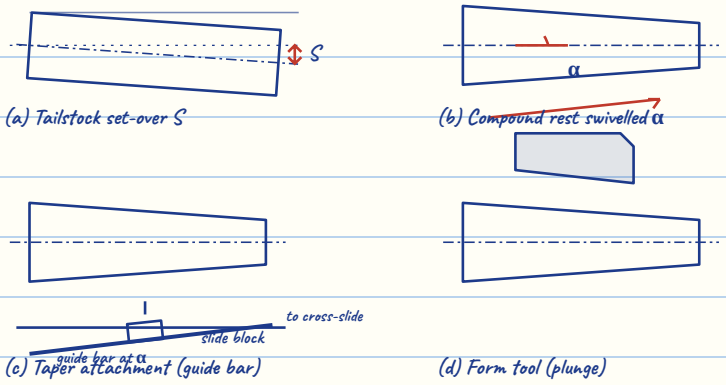


Fig 4: Taper turning methods

> ★ **Important**
>
> - Tailstock set-over: $S = L(D-d)/2l$; Attachment: best for long accurate tapers ; Compound rest: steep / short
> - Taper turn: tool tip must be at centre height

8. Thread Cutting on Lathe

- Work rotates, tool moves a distance = **pitch** per revolution → helical groove
- Tool: 60° (V, metric/Whitworth 55°) single point, ground to thread form, set **square** with a centre gauge
- Spindle → **change gears** (quadrant) → lead screw → **half nut** → carriage

Gear ratio = Driver / Driven = Pitch of job (P) / Lead of lead screw (L)

$$Z_{driver} / Z_{driven} = P / L = (N_{lead\ screw}) / (N_{spindle})$$

- Simple gearing: 1 driver (on spindle stud), 1 driven (on lead screw) + idler
- **Compound gearing** when ratio is large: (A/B) × (C/D) = P/L
- Eg: cut 3 mm pitch, lead screw 6 mm: ratio = 3/6 = 1/2 → 20T driver : 40T driven (or 30/60)
- Eg: cut 2 mm pitch, L.S. 6 mm → 1/3 → (20/40)×(30/45) ... any combination equal to 1/3
- Inch L.S. (4 TPI) to cut 10 TPI: ratio = 4/10 = 2/5 → 40T/100T (or ×20 = 40/100)

Half nut & thread chasing dial

- **Half nut** – 2 halves clamp on lead screw by lever; engaged for threads, disengaged for others
- Cut passes: tool returned after each pass (reverse spindle or withdraw), re-engage half nut for next pass
- **Thread chasing (indicator) dial** – small dial geared to lead screw on apron; shows when to engage half nut at same point so tool follows same groove
- Even no. of threads in L.S./job → engage at any line; odd → at numbered lines; fractional → specific line / keep half nut engaged
- Steps: ① set speed (slow) & gears ② set tool square, centre height ③ touch, zero cross-slide ④ light cut, check pitch with gauge ⑤ cut passes with compound rest set 29° (feeds on one flank) ⑥ finish & check by thread gauge
- Depth of V-thread ≈ 0.6495 P (ISO ≈ 0.6134 P) ; Whitworth ≈ 0.64 P

Lead = distance moved by nut in 1 rev (= pitch × starts). For multi-start thread, lead = n × pitch.

9. Lathe Attachments

- **Taper turning attachment** – long accurate tapers
- **Steady & follow rests** – support slender work
- **Milling attachment** – on cross slide, milling keyways
- **Grinding attachment** – tool-post grinder, finishes hard work
- **Copying (tracer) attachment** – hydraulic, copies template profile
- **Gear cutting attachment**, **Relieving attachment** (milling cutters, taps)

- Thread chasing dial, Quick-change gear box, Boring attachment
- Turret / multi-tool post, Cam / Eccentric turning devices, Collet attachment

10. Machining Time – Lathe

$$N = 1000\,V / (\pi D) \text{ rpm} \quad (V \text{ m/min}, D \text{ mm})$$

Turning

$$T = L / (f N) \text{ min}, \text{ with } i \text{ passes: } T = i \cdot L / (f N)$$

- L = length of cut + approach + over-travel (mm) ; f = feed (mm/rev) ; N rpm
- Number of passes i = total depth / depth per cut
- Feed per min = $f N$; MRR = $\pi D_{avg} f d V$ (mm^3/min)

Facing

$$T = (D/2) / (f N) = D / (2 f N)$$

- For constant V, N increases as D falls (CSS in CNC)

Drilling on lathe / drill press

$$T = (L + A) / (f N) \; ; \; A = 0.3\,d \text{ (approach) } [118^\circ \text{ drill: } A = 0.5\,d \cot 59^\circ \approx 0.3\,d]$$

Thread cutting & parting

$$T = i \cdot L / (P N) \text{ (feed = pitch } P \text{ per rev)}$$

Ex 1 (turning). Turn 50 mm dia, 200 mm long, V = 30 m/min, f = 0.25 mm/rev, one pass.
$N = 1000 \times 30/(\pi \times 50) =$ **191 rpm** ; $T = 200 / (0.25 \times 191) =$ **4.19 min**

Ex 2 (facing). Face 100 mm dia bar, N = 200 rpm, f = 0.2 mm/rev.
$T = 100 / (2 \times 0.2 \times 200) =$ **1.25 min**

Ex 3 (taper). Taper job D = 60, d = 40, l = 200 = full length (L = 200).
$\tan\alpha = (60-40)/(2 \times 200) = 0.05 \rightarrow \alpha =$ **2.86°** ; set-over $S =$ $200 \times 20/(2 \times 200) =$ **10 mm**

Ex 4 (threading). 2 mm pitch, 40 mm long, N = 60 rpm, 8 passes.
$T = 8 \times 40/(2 \times 60) =$ **2.67 min**

11. Drilling Machines

Drilling = making a round hole with a rotating **twist drill** (rotation + axial feed). Drill rotates, work fixed. Also gives centre drilling, reaming etc. by changing tool.

Types

- **Portable / hand drill** – electric, for site work
- **Sensitive (bench) drilling m/c** – small, up to 12–15 mm; hand feed gives "feel"; high speed
- **Upright (pillar) drill** – round column; table adjustable; medium jobs, power feed; 50 mm dia
- **Radial drilling m/c** – radial arm swings about column & moves up/down; drill head slides on arm; large & heavy jobs, hole position without moving job
- **Gang drilling m/c** – 2–6 spindles on one long table, each with different tool; sequence ops (drill, ream, tap) ; batch
- **Multi-spindle m/c** – many spindles with one head, drill many holes at once at fixed positions; mass production (auto parts)
- **Deep-hole drilling m/c** – L/D > 10 ; gun drill / BTA; high-pressure coolant through drill; horizontal; barrels, crankshafts
- **Turret / automatic / CNC drilling** m/c

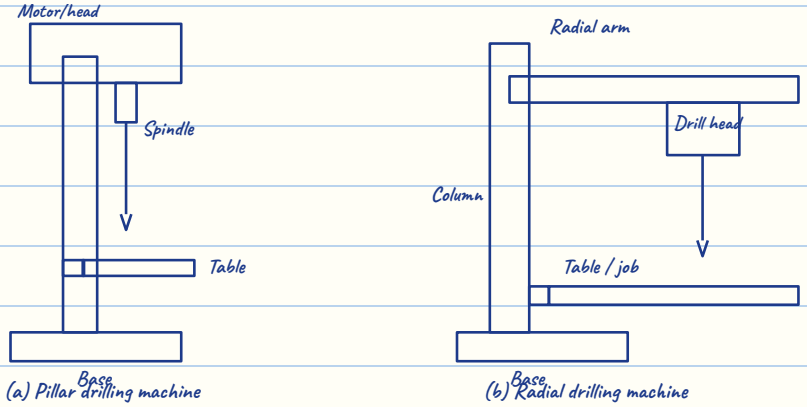


Fig 6: Pillar and radial drilling machines

Specification

- Max drill dia in steel ; table size ; spindle travel ; no. of speeds/feeds ; Morse taper no. ; swing (= 2 × distance column to spindle) ; power; radial: arm length, column dia

Twist drill nomenclature (Fig 7)

- **Point angle** (118° general; 135° hard/steel) ; **lip clearance** 8-12° ; **helix angle** 24-32° (rake)
- **Chisel edge angle** 115-135° ; **web** (thickens to shank) ; **flutes** (chip removal, coolant)

- **Land**, **margin** (guides), **body clearance**, **lips** (cutting edges), **shank** (straight / Morse taper), **tang** (drives drill)

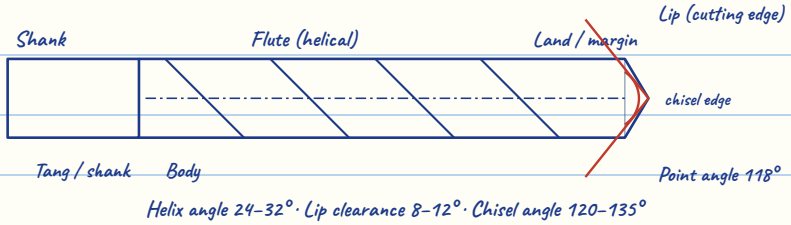


Fig 7: Twist drill – nomenclature

Tool & work holding (drilling)

- **Drill chuck** (jaws, key) for straight shank ; **sleeve / socket** & **drift** for Morse taper shank
- Work: **machine vice, V-block, T-bolts & clamps, angle plate, jig**

Operations on drilling m/c (Fig 8)

Operation	Meaning
Drilling	Make new hole
Reaming	Finish hole to accurate size, good finish (multi-tooth reamer, small allowance 0.2–0.5 mm)
Boring	Enlarge / correct hole by single-point bar
Counter-boring	Enlarge end of hole, square shoulder (for bolt heads)
Counter-sinking	Conical enlargement for flat-head screw (60°/90°)
Spot-facing	Smooth flat seat around hole for nut/washer
Tapping	Cut internal threads with tap
Trepanning / Centring	Cut annular groove for big holes ; centre drill

Drilling time

$$N = 1000\,V / (\pi\, d)\ ;\ T = (L + A) / (f\, N)\ ;\ A \approx 0.3\, d$$

- L = depth of hole (for blind); through hole: L = thickness + approach + breakthrough (≈ 1–2 mm)
- $MRR = (\pi\, d^2/4)\, f\, N$

Ex 5. Drill 20 mm hole, 40 mm deep, V = 25 m/min, f = 0.2 mm/rev.
$N = 1000\times25/(\pi\times20) = 398$ rpm ; $A = 0.3\times20 = 6$ mm ; $T = (40+6)/(0.2\times398) =$ **0.58 min**

12. Boring Machines

Boring = enlarging & finishing an existing hole with **single point** tool on a boring bar (concentric, straight, accurate).

- **Horizontal boring m/c (HBM)** – horizontal spindle; table (X, Z), head moves on column (Y); outboard support; big castings, gear boxes, engine blocks; also drilling, milling, facing
  - Types: table, planer, floor, multiple-head
- **Vertical boring m/c (VBM / vertical turret lathe)** – vertical spindle, **rotating table** (like inverted lathe); heavy, large dia, short work (flywheels, rings, turbine casings); cross rail, 2 heads
- **Jig boring m/c** – very high precision (±2 μm); coordinate positioning by micro-scales; temp. controlled room; jigs, fixtures, dies
- **Precision boring** – diamond / carbide tools, high speed light cut; engine cylinders, bearings

Boring tools

- Single-point bit in **boring bar** (set screw) ; **offset boring head** (micro adjust dia) ; **line boring** (several coaxial holes) ; **back boring** ; **floating** reamer type tool
- Bar overhang short; $L/D \leq 4$ for steel bar (carbide bars up to 7)

Drilling	Boring
Makes new hole	Enlarges existing hole
Multi-edge twist drill	Single-point tool
Poor concentricity	Corrects position / alignment

13. Shaper

Shaper = reciprocating m/c; **single-point tool** moves to-and-fro (cutting on forward stroke), work fixed on table gets **intermittent feed** at the end of return stroke. Produces flat (horizontal, vertical, inclined), grooves, slots, gear teeth.

Parts

- **Base** – C.I., supports all
- **Column** – houses drive; guideways on top for ram
- **Ram** – reciprocates in dovetail guides, carries tool head
- **Tool head** – **down feed screw, swivel base (angular cuts), apron with clapper box, tool post**
- **Clapper box** – tool lifts on return stroke (no drag)
- **Cross-rail** (elevates), **saddle** (cross feed), **table** (T-slots, work vice)

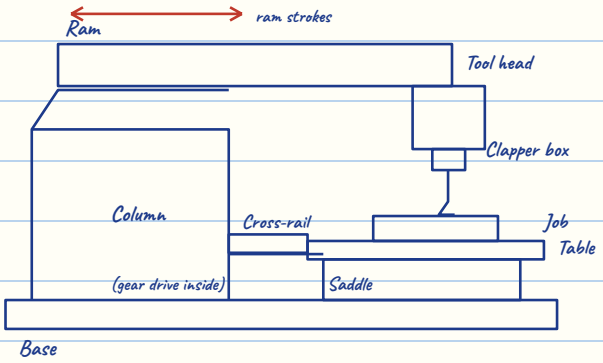


Fig 9: Shaper – schematic with main parts

Quick-return mechanisms (Fig 10)

- (a) **Crank & slotted link** (most common): bull gear + sliding block (crank pin) → slotted link (rocker arm) pivoted at bottom → ram via link
- (b) **Whitworth QR** – crank centre inside slotted lever pivot circle; used in heavy shapers/slotters; ratio fixed by geometry
- (c) **Hydraulic shaper** – oil piston, constant cutting speed, smooth, stepless speed, overload safe

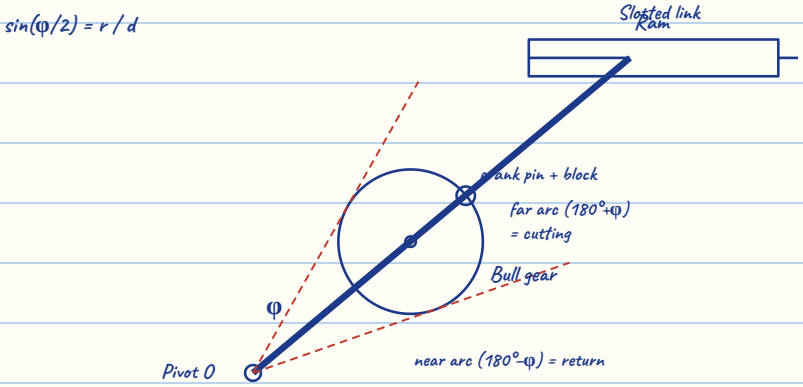


Fig 10: Crank & slotted-link quick-return mechanism (O to bull-gear centre = d, crank radius = r)

- Cutting stroke: crank turns through angle $(180° + \varphi)$; return: $(180° - \varphi)$ → return faster at const. bull-gear speed
- Return ratio: $\sin(\varphi/2)$ = crank radius / distance between centres

$$\text{Quick return ratio} = \text{cutting time / return time} = (180 + \varphi) / (180 - \varphi) \quad (> 1, \approx 3{:}2)$$

Specification & types

- Max length of stroke (150–900 mm) ; table size ; max vertical travel ; power ; no. of strokes/min
- **Types:** crank / geared / hydraulic (drive) ; horizontal, vertical (slotter), travelling head (long jobs) ; push-cut / draw-cut ; standard / universal (table tilts)

Work holding & tools

- Machine vice, T-bolts & clamps, V-blocks, angle plate, special fixtures, stop pins
- Tools: round-nose roughing, finishing broad tool, side-recess, parting, form tool ; HSS ; relief for return stroke

Operations

- Horizontal, vertical, angular surfaces ; **slots, grooves, keyways, T-slots, dovetails** ; irregular / contour ; gear teeth (rack) ; splines

Machining time

- Length of stroke L = job length + 20–30 mm clearance (approach + overtravel)
- Let m = return time / cutting time = 1/(ratio) , V = cutting speed (m/min)

$$\text{Strokes per min } N = 1000\,V / [\,L\,(1+m)\,] \quad (L \text{ in mm})$$

$$T = W / (f \times N) \text{ min} \quad (W = \text{width} + \text{clearance},\ f = \text{feed mm/stroke})$$

- Number of cuts i = total depth / depth per cut → multiply
- Or time/stroke = $L/V_c + L/V_r$

Ex 6. Shape plate 300 mm wide, 180 mm long ; clearance 20 → L = 200 mm. V = 15 m/min, ratio 3 : 2 (m = 2/3), f = 1.5 mm/stroke.
$N = 1000\times15/[200\times(1+0.667)]$ = **45 strokes/min** ; $T = 300 / (1.5\times45)$ = **4.44 min**

★ *Important*

- Cutting on forward stroke only ; feed at end of return stroke ; ram stroke adjusted by crank-pin radius (position by ram screw)
- Ram speed variable along stroke (max at mid) ; ratio of return ≈ 2:3

14. Slotter (Slotting Machine)

Slotter = vertical shaper; ram moves **vertically**, cutting on **downward** stroke; tool is single-point; work on rotary table with **3 feeds**: longitudinal, cross, rotary.

- **Parts:** base, column (hollow, houses drive), saddle, cross-slide, **circular (rotary) table**, ram + tool head, drive (Whitworth QR or hydraulic)
- Ram can be tilted slightly ($\leq 10°$) for tapered cuts
- **Types:** puncher (heavy), precision (tool-room), tool-room slotter
- **Operations:** internal keyways, splines, slots, internal gears, circular & contour surfaces, square / hexagonal holes

- Whitworth mechanism: crank pin rotates; slotted-link end (pivot at centre of gear) ; ram moves down slow (cut), up fast

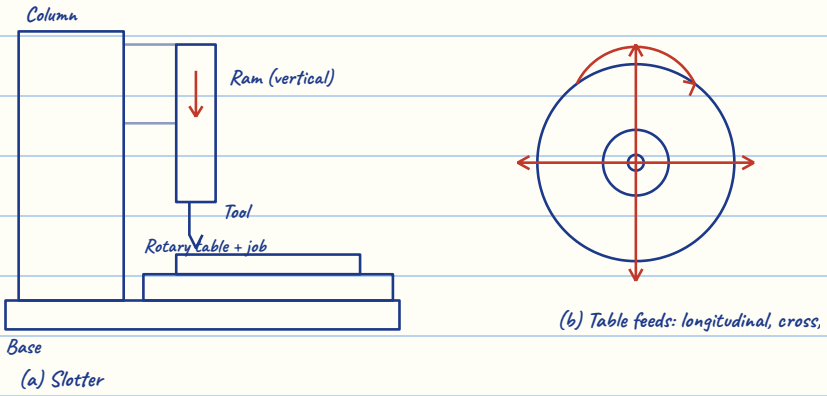


Fig 11: Slotting machine and its three table feeds (top view of table)

15. Planer (Planing Machine)

Planer = for large, heavy flat surfaces. **Work reciprocates** on table; tool fixed (feeds intermittently across at the start of the return stroke). Cutting on forward stroke, tool lifts on return (clapper box).

Parts

- **Bed** (long, with V & flat ways) ; **Table** (T-slots, holds work, moves on bed)
- **Housings / columns** (one or two, vertical) ; **Cross-rail** (elevating, carries tool heads)
- **Tool heads** – 2 on rail, 1–2 on housings (side heads) ; each with clapper box, tool post, swivel
- **Driving mechanisms** (drive) ; **Feed** mechanism ; reversing dogs

Drives of planer

- **Open and cross belt drive** – 2 belts, forward (open, slow) & return (cross, fast); ratio 1:3–1:4 ; old
- **Reversible motor drive** – D.C. motor via gears & rack-pinion; dogs reverse the motor; Ward-Leonard control
- **Hydraulic drive** – piston, smooth, speed constant, high loads
- **Crank drive** – small planers
- Table speed: cutting 6–30 m/min ; return 2–4 × cutting

Types

- **Double-housing** – two housings joined by cross-rail ; rigid ; width of work limited by housing gap ; most common
- **Open-side (single-housing)** – one housing, overhanging cross-rail ; very wide work ; less rigid

- **Pit planer** – tool bridge moves over fixed table in pit ; very large & heavy jobs
- **Edge (plate) planer** – plate edges for welding preparation (bevel) ; work clamped & tool moves along edge
- **Divided-table planer** – 2 tables, setting up one while other cuts

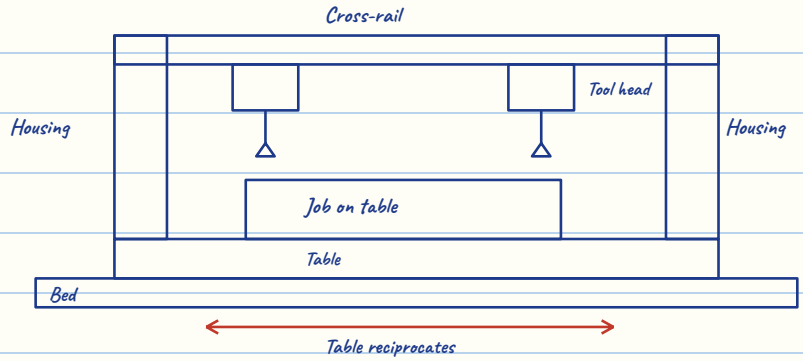


Fig 12: Double-housing planer

Specification

- Max length of table travel ; max width between housings ; max height under rail ; table size ; cutting & return speeds ; no. of tool heads ; motor kW

Tools, work holding, operations

- Tools: like shaper but heavier, with larger cross-section ; carbide-tipped, broad finishing tool
- Holding: T-bolts, clamps, **planer centres**, stops, angle plates, toe-dogs, jacks
- Ops: flat, horizontal / vertical / inclined surfaces, slots, T-slots, dovetails, grooves – machine beds, tables, large castings, ways of lathes ; several jobs simultaneously

Machining time (planer)

$$T = (W / f) \times [\, L / V_c + L / V_r \,]$$ (L = table travel (job + overtravel), V in m/min)

Ex 7. Planer job 1000 mm long, 600 mm wide ; overtravel 100 mm → $L = 1.1$ m ; $V_c = 12$ m/min, $V_r = 36$ m/min, $f = 2$ mm/stroke.

Time/double stroke = $1.1/12 + 1.1/36 = 0.0917 + 0.0306 = 0.1222$ min ; strokes = $600/2 = 300$

$T = 300 \times 0.1222 =$ **36.7 min**

16. Comparison: Shaper vs Planer vs Slotter

Point	Shaper	Planer	Slotter
Moving part	Tool (ram)	Work (table)	Tool (vertical ram)
Job size	Small / medium	Large, heavy	Medium (internal)
Cutting stroke	Forward, horizontal	Table forward, horiz.	Downward, vertical
Feed	Table, at end of return	Tool, at start of return	Longit., cross, rotary
Tool heads	1	2–4	1
Drive	Crank-slotted link / hydraulic	Reversing motor / belt / hydraulic	Whitworth / hydraulic
Rigidity, cut	Light cuts	Heavy, accurate	Light / medium
Use	Flat, slots, keyways	Beds, tables, ways	Internal keyways, gears
Cost, floor	Low	High, large floor	Medium

★ **Important**

- Shaper: tool moves, work feeds ; Planer: work moves, tool feeds
- Mnemonic: **S**haper = **S**mall, **P**laner = **P**onderous, **S**lotter = **S**tanding (vertical)
- Quick return ratio for shaper ≈ 3:2 (cutting : return)

17. Quick Revision & Likely Exam Questions

Formulas

$N = 1000V/\pi D$; $T(\text{turn}) = L/fN$; $T(\text{face}) = D/2fN$; $T(\text{drill}) = (L+0.3d)/fN$; $\tan\alpha = (D-d)/2l$; $QRR = (180+\varphi)/(180-\varphi)$

Likely questions

1. Sketch an engine lathe & explain its main parts. (long)
2. Specify & classify lathes ; lathe operations & attachments.
3. Describe work holding devices (chucks, centres, mandrels, rests) used on lathe.
4. Explain taper turning methods ; derive tailstock set-over.
5. Explain thread cutting on lathe – change gears, half nut, chasing dial.
6. Derive machining time for turning, facing & drilling; solve numerical.
7. Classify drilling machines ; radial drill sketch ; twist drill nomenclature.
8. Horizontal, vertical & jig boring m/c ; boring tools.
9. Shaper : parts, quick-return mechanism, machining time.
10. Slotter vs shaper ; planer types & drives.